MLH HACK DAY 2026

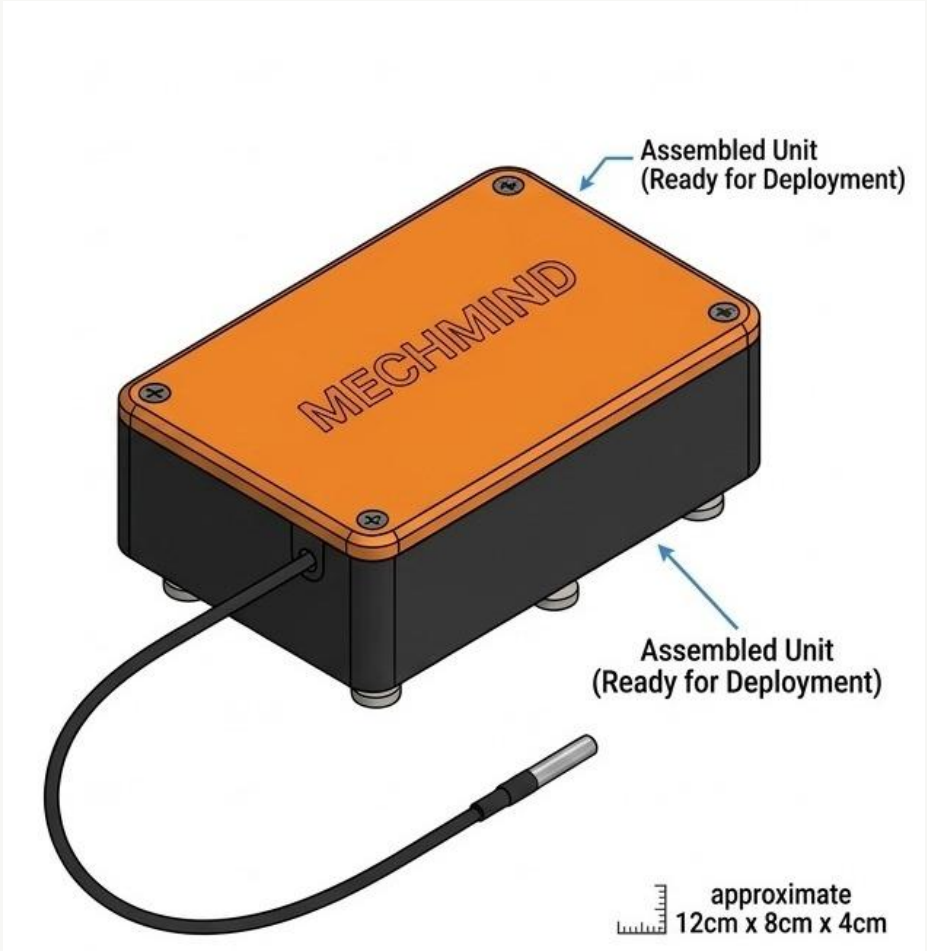
MechMind AI

Powered by Google Gemma 4

On DigitalOcean

An edge-first Gemma 4 copilot, fed by a $25 magnetic sensor node, that diagnoses heavy machinery over WhatsApp: offline in the field, scaled in the cloud.

Oni Mujeeb Ramadan · Mubarak Raji Babatunde · Olawale Oladele
Lagos State University



github.com/DevMubarak1/mechmind-ai · MIT License

THE PROBLEM

One breakdown can strand a machine for weeks

$1.2B

lost every year to heavy machinery downtime across Africa

Rural mechanics have no scan tools, so minor sensor faults get misdiagnosed as engine replacements.

$15,000

for an OEM diagnostic tool such as Cat ET

Caterpillar, John Deere and Komatsu lock diagnostics behind proprietary hardware.

2–3 weeks

is the whole planting or harvest window

One failed hydraulic pump or bearing can spoil a harvest. Farm sites often have only 2G/3G, or nothing.

HOW MECHMIND WORKS

Gemma 4 is the brain of the system

ESP32 NODE-001

Vibration, temperature and sound features every 2 s

Mechanic on WhatsApp

Symptoms in their own words, or a photo of the damage

FASTAPI BACKEND · DEPLOYS TO DIGITALOCEAN

ChromaDB RAG

1,200+ pages of Cat, Komatsu, Perkins and XCMG manuals

Safety code pinning

Critical SAE J1939 codes get verified shutdown steps

Google Gemma 4

Edge: gemma4:e2b on Ollama · Cloud: Gemini API

WhatsApp copilot

Cause, tools and repair steps in about 3 s

Fleet dashboard

Live gauges, SSE alerts and copilot chat



THE SENSOR NODE

Gemma’s eyes and ears

ESP32-WROOM-32	Dual-core 240 MHz MCU with Wi-Fi/BLE. Runs the edge signal processing.
ADXL345 accelerometer	3-axis vibration at 800 Hz: RMS and peak-to-peak for unbalance and bearing wear.
DS18B20 temperature probe	Waterproof, ±0.5 °C. Tracks rate of rise to catch overheating before seizure.
INMP441 mic + 18650 cell	Acoustic cavitation detection. 2600 mAh Li-ion with TP4056 charging, 48+ hours in the field.

Neodymium magnet mount · 25 kg pull · attaches in 30 s

LIVE DEMO 1

Live telemetry gives Gemma 4 real context

2 s

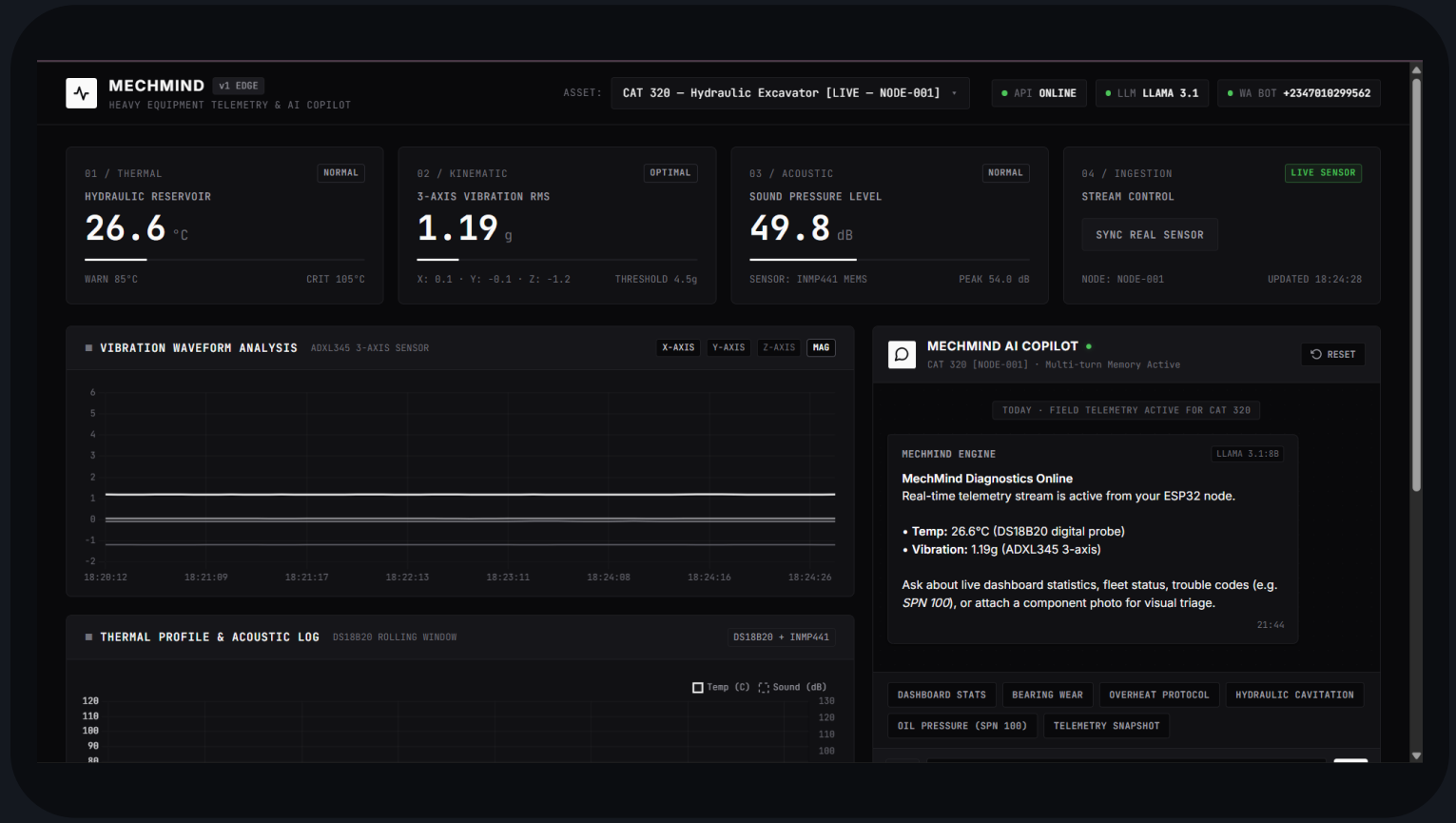
between readings, so Gemma always answers from the machine's current state

99.88%

less cellular data: 1.6 MB/min of raw samples becomes 3.6 KB/min

< 1 s

to push critical alerts to the screen over Server-Sent Events



GROUNDED AND SAFE

Gemma 4 answers only from OEM manuals

Alert or question

A telemetry anomaly fires, or a mechanic texts a symptom or photo.

Retrieve

ChromaDB searches 1,200+ manual pages, split into 500-token chunks.

Pin safety

Critical SAE J1939 codes (e.g. SPN 100 FMI 1, low oil pressure) put shutdown steps first.

Gemma 4

Writes the plain-language answer, grounded only in what was retrieved.

75.0%

benchmark diagnostic accuracy

0.0%

safety hallucination rate

1,200+

pages of OEM service manuals

GEMINI API + DIGITALOCEAN

Edge, cloud and DigitalOcean: one Gemma 4 engine

EDGE · IN THE FIELD

gemma4:e2b

- Runs on Ollama, offline
- No recurring API charges
- Answers the mechanic on site

CLOUD · ACROSS THE FLEET

Gemini API

- Fleet-wide batch processing
- High-throughput workloads
- Same grounded prompts

DEPLOY · DIGITALOCEAN

App Platform

- digitalocean-app.yaml
- setup_droplet.sh: live in < 2 min
- Hermes + GPU Droplets

Under $2/month per machine to monitor a fleet on DigitalOcean.

LIVE DEMO 2 · FIELD DELIVERY

Gemma 4 answers right inside WhatsApp

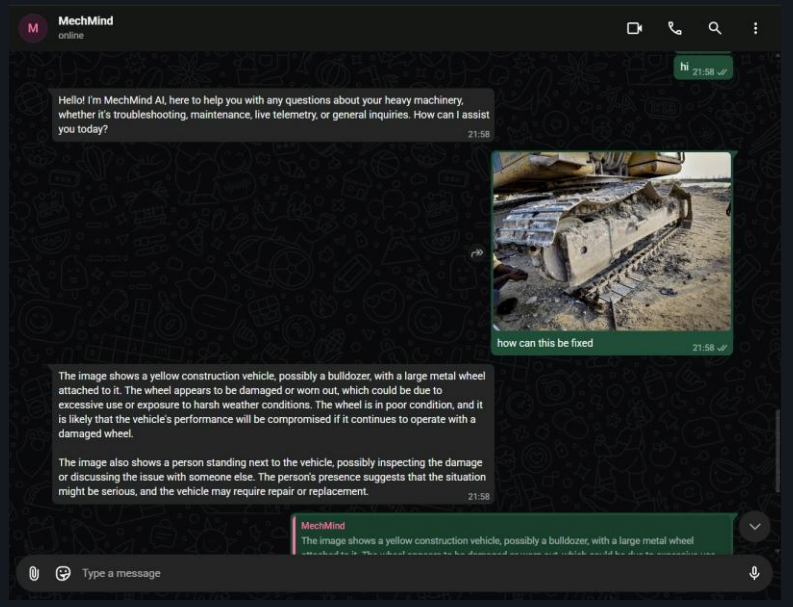
95% of African mechanics already use WhatsApp daily. No app, no login.

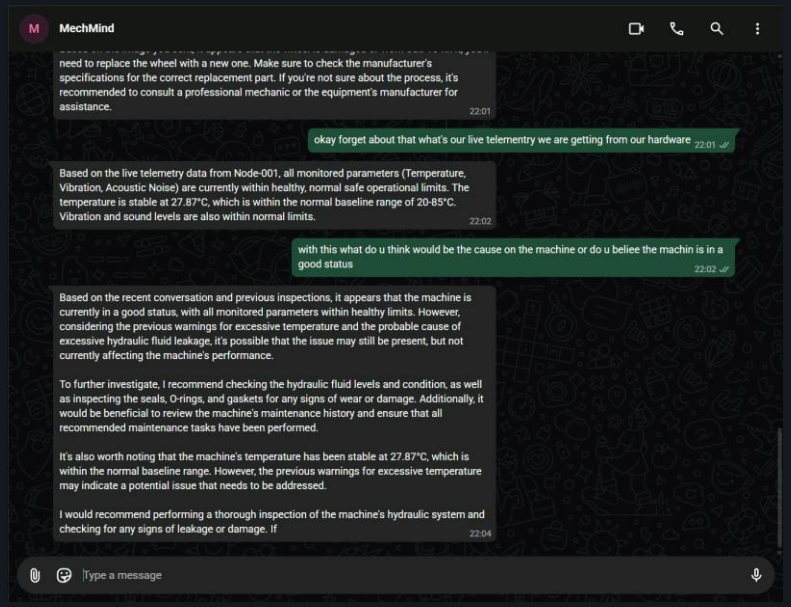
Text a symptom

“Hydraulic pump makes a high-pitched whine and temp is 88 °C”: cause, wrench sizes and steps in about 3 s, using NODE-001’s live readings.

Send a photo

Gemma 4 reads images too: snap the damaged part and get a diagnosis and next steps.





Gemma 4 puts a diagnostic expert in every mechanic's pocket

Oni Mujeeb Ramadan

Civil Engineering, LASU

Product operations, outreach and field validation

Mubarak Raji Babatunde

Civil Engineering, LASU

AI lead: Gemma 4 RAG, FastAPI, DigitalOcean

Olawale Oladele

Mechanical Engineering, LASU

Hardware lead: ESP32 firmware, sensors, CAD

OPEN SOURCE · MIT LICENSE

Gemma 4 · ChromaDB · Baileys WhatsApp gateway · ESP32 firmware · Next.js + FastAPI · DigitalOcean deploy

Competing for: Leverage Gemma 4 through Google Gemini APIs · DigitalOcean Open-Source AI · Best Open-Source AI Project

github.com/DevMubarak1/mechmind-ai · mechmind-ng.vercel.app

Thank you